laptop data

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In [83]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
```

In [84]:

```
data = pd.read_csv('laptopData.csv')
```

In [85]:

```
data.head(5)
```

Out [85]:

	Unnamed: 0	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	
0	0.0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x…	Intel Core i5 2.3GHz	8GB	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37kg	71378.6
1	1.0	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8GB	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34kg	47895.5
2	2.0	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8GB	256GB SSD	Intel HD Graphics 620	No OS	1.86kg	30636.0
3	3.0	Apple	Ultrabook	15.4	IPS Panel Retina Display 2880x…	Intel Core i7 2.7GHz	16GB	512GB SSD	AMD Radeon Pro 455	macOS	1.83kg	135195
4	4.0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x…	Intel Core i5 3.1GHz	8GB	256GB SSD	Intel Iris Plus Graphics 650	macOS	1.37kg	96095.8

In [86]:

```
data.shape
```

Out [86]:

```
(1303, 12)
```

In [87]:

```
data.info()
```

Out [87]:

```
<class 'pandas.DataFrame'>
RangeIndex: 1303 entries, 0 to 1302
Data columns (total 12 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Unnamed: 0        1273 non-null   float64
 1   Company           1273 non-null   str
 2   TypeName          1273 non-null   str
 3   Inches            1273 non-null   str
 4   ScreenResolution  1273 non-null   str
 5   Cpu               1273 non-null   str
 6   Ram               1273 non-null   str
 7   Memory            1273 non-null   str
 8   Gpu               1273 non-null   str
 9   OpSys             1273 non-null   str
 10  Weight            1273 non-null   str
 11  Price             1273 non-null   float64
dtypes: float64(2), str(10)
memory usage: 122.3 KB
```

In [88]:

```
data.drop(['Unnamed: 0'], axis=1, inplace=True)
```

In [89]:

```
data.head()
```

Out [89]:

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price
0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x…	Intel Core i5 2.3GHz	8GB	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37kg	71378.6832
1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8GB	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34kg	47895.5232
2	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8GB	256GB SSD	Intel HD Graphics 620	No OS	1.86kg	30636.0000
3	Apple	Ultrabook	15.4	IPS Panel Retina Display 2880x…	Intel Core i7 2.7GHz	16GB	512GB SSD	AMD Radeon Pro 455	macOS	1.83kg	135195.3360
4	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x…	Intel Core i5 3.1GHz	8GB	256GB SSD	Intel Iris Plus Graphics 650	macOS	1.37kg	96095.8080

In [90]:

```
data.drop(['ScreenResolution'], axis=1, inplace=True)
```

In [91]:

```
data.drop(['TypeName'], axis=1, inplace=True)
```

In [92]:

```
data.info()
```

Out [92]:

```
<class 'pandas.DataFrame'>
RangeIndex: 1303 entries, 0 to 1302
Data columns (total 9 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   Company  1273 non-null   str
 1   Inches   1273 non-null   str
 2   Cpu      1273 non-null   str
 3   Ram      1273 non-null   str
 4   Memory   1273 non-null   str
 5   Gpu      1273 non-null   str
 6   OpSys    1273 non-null   str
 7   Weight   1273 non-null   str
 8   Price    1273 non-null   float64
dtypes: float64(1), str(8)
memory usage: 91.7 KB
```

In [93]:

```
data.shape
```

Out [93]:

```
(1303, 9)
```

In [94]:

```
data['Company'].isnull().sum()
```

Out [94]:

```
np.int64(30)
```

In [95]:

```
data['Company'].apply(lambda x:  'macOS' if x == 'apple' else ('Windows' if x == 'hp' else 'Windows'))
```

Out [95]:

```
0       Windows
1       Windows
2       Windows
3       Windows
4       Windows
         ...
1298    Windows
1299    Windows
1300    Windows
1301    Windows
1302    Windows
Name: Company, Length: 1303, dtype: str
```

In [96]:

```
data['OpSys'].apply(lambda x:  'Apple' if x == 'macOS' else 'win os')
```

Out [96]:

```
0        Apple
1        Apple
2       win os
3        Apple
4        Apple
         ...
1298    win os
1299    win os
1300    win os
1301    win os
1302    win os
Name: OpSys, Length: 1303, dtype: str
```

In [97]:

```
data = data.dropna(subset=['Ram'])
```

In [98]:

```
data['Ram'].isnull().sum()
```

Out [98]:

```
np.int64(0)
```

In [99]:

```
data['Ram'] = data['Ram'].str.replace('GB', '').astype(int)
```

In [100]:

```
data['Memory'] = data['Memory'].str.replace('SSD', '')
```

In [101]:

```
data['Memory'] = data['Memory'].str.replace('GB', '')
```

In [102]:

```
data['Memory(GB)'] = data["Memory"]
```

In [103]:

```
data['Ram(GB)'] = data['Ram']
```

In [104]:

```
data.drop(['Memory', 'Ram'], axis=1, inplace=True)
```

In [105]:

```
data['Price'].astype(int)
```

Out [105]:

```
0        71378
1        47895
2        30636
3       135195
4        96095
         ...
1298     33992
1299     79866
1300     12201
1301     40705
1302     19660
Name: Price, Length: 1273, dtype: int64
```

In [106]:

```
data['Weight(KG)'] = data['Weight'].str.replace('kg', '')
```

In [107]:

```
data.drop(['Weight'], axis=1, inplace=True)
```

In [108]:

```
data['OpSys'].value_counts()
```

Out [108]:

```
OpSys
Windows 10      1047
No OS             63
Linux             61
Windows 7         45
Chrome OS         27
macOS             13
Mac OS X           8
Windows 10 S       8
Android            1
Name: count, dtype: int64
```

In [109]:

```
data['Memory(GB)'] = data['Memory(GB)'].str.replace('Flash Storage', '')
```

In [110]:

```
data['Memory(GB)'] = data['Memory(GB)'].str.replace('Hybrid', '')
```

In [111]:

```
data['Memory(GB)'] = data['Memory(GB)'].str.replace('HDD', '')
```

In [112]:

```
data['Memory(GB)'].value_counts()
```

Out [112]:

```
Memory(GB)
256               409
1TB               230
500               130
512               118
128  +  1TB        92
128                79
256  +  1TB        71
32                 44
2TB                16
64                 15
512  +  1TB        14
256  +  2TB        10
1.0TB              10
16                 10
180                 4
512  +  2TB         3
1TB  +  1TB         3
256  +  256         2
256  +  500         2
128  +  2TB         2
512  +  512         1
512  +  256         1
64  +  1TB          1
?                   1
240                 1
8                   1
508                 1
512  +  1.0TB       1
256  +  1.0TB       1
Name: count, dtype: int64
```

In [124]:

```
data['Memory(GB)'] = data['Memory(GB)'].str.replace('TB', '000')
```

```
data['SSD'] = data['Memory(GB)'].str.extract(r'(\d+)\s*(?:SSD|Flash|$)')
data['HDD'] = data['Memory(GB)'].str.extract(r'(\d+)\s*HDD')

data['SSD'] = data['SSD'].fillna(0).astype(int)
data['HDD'] = data['HDD'].fillna(0).astype(int)
```

In [127]:

```
data.drop(["HDD"],axis=1,inplace=True)
```

In [114]:

```
data['Memory(GB)'].value_counts()
```

Out [114]:

```
Memory(GB)
256                 409
1000                230
500                 130
512                 118
128  +  1000         92
128                  79
256  +  1000         71
32                   44
2000                 16
64                   15
512  +  1000         14
256  +  2000         10
1.0000               10
16                   10
180                   4
512  +  2000          3
1000  +  1000         3
256  +  256           2
256  +  500           2
128  +  2000          2
512  +  512           1
512  +  256           1
64  +  1000           1
?                     1
240                   1
8                     1
508                   1
512  +  1.0000        1
256  +  1.0000        1
Name: count, dtype: int64
```

In [115]:

```
data.head(5)
```

Out [115]:

	Company	Inches	Cpu	Gpu	OpSys	Price	Memory(GB)	Ram(GB)	Weight(KG)	SSD	HDD
0	Apple	13.3	Intel Core i5 2.3GHz	Intel Iris Plus Graphics 640	macOS	71378.6832	128	8	1.37	128	0
1	Apple	13.3	Intel Core i5 1.8GHz	Intel HD Graphics 6000	macOS	47895.5232	128	8	1.34	128	0
2	HP	15.6	Intel Core i5 7200U 2.5GHz	Intel HD Graphics 620	No OS	30636.0000	256	8	1.86	256	0
3	Apple	15.4	Intel Core i7 2.7GHz	AMD Radeon Pro 455	macOS	135195.3360	512	16	1.83	512	0
4	Apple	13.3	Intel Core i5 3.1GHz	Intel Iris Plus Graphics 650	macOS	96095.8080	256	8	1.37	256	0

In [116]:

```
data.info()
```

Out [116]:

```
<class 'pandas.DataFrame'>
Index: 1273 entries, 0 to 1302
Data columns (total 11 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Company     1273 non-null   str
 1   Inches      1273 non-null   str
 2   Cpu         1273 non-null   str
 3   Gpu         1273 non-null   str
```

```
 4   OpSys       1273 non-null   str
 5   Price       1273 non-null   float64
 6   Memory(GB)  1273 non-null   str
 7   Ram(GB)     1273 non-null   int64
 8   Weight(KG)  1273 non-null   str
 9   SSD         1273 non-null   int64
 10  HDD         1273 non-null   int64
dtypes: float64(1), int64(3), str(7)
memory usage: 119.3 KB
```

In [117]:

```
data.shape
```

Out [117]:

```
(1273, 11)
```

In [118]:

```
data.head()
```

Out [118]:

	Company	Inches	Cpu	Gpu	OpSys	Price	Memory(GB)	Ram(GB)	Weight(KG)	SSD	HDD
0	Apple	13.3	Intel Core i5 2.3GHz	Intel Iris Plus Graphics 640	macOS	71378.6832	128	8	1.37	128	0
1	Apple	13.3	Intel Core i5 1.8GHz	Intel HD Graphics 6000	macOS	47895.5232	128	8	1.34	128	0
2	HP	15.6	Intel Core i5 7200U 2.5GHz	Intel HD Graphics 620	No OS	30636.0000	256	8	1.86	256	0
3	Apple	15.4	Intel Core i7 2.7GHz	AMD Radeon Pro 455	macOS	135195.3360	512	16	1.83	512	0
4	Apple	13.3	Intel Core i5 3.1GHz	Intel Iris Plus Graphics 650	macOS	96095.8080	256	8	1.37	256	0

In [119]:

```
data['Price']= data['Price'].astype(int)
```

In [120]:

```
data.head()
```

Out [120]:

	Company	Inches	Cpu	Gpu	OpSys	Price	Memory(GB)	Ram(GB)	Weight(KG)	SSD	HDD
0	Apple	13.3	Intel Core i5 2.3GHz	Intel Iris Plus Graphics 640	macOS	71378	128	8	1.37	128	0
1	Apple	13.3	Intel Core i5 1.8GHz	Intel HD Graphics 6000	macOS	47895	128	8	1.34	128	0
2	HP	15.6	Intel Core i5 7200U 2.5GHz	Intel HD Graphics 620	No OS	30636	256	8	1.86	256	0
3	Apple	15.4	Intel Core i7 2.7GHz	AMD Radeon Pro 455	macOS	135195	512	16	1.83	512	0
4	Apple	13.3	Intel Core i5 3.1GHz	Intel Iris Plus Graphics 650	macOS	96095	256	8	1.37	256	0

In [121]:

```
data.head()
```

Out [121]:

	Company	Inches	Cpu	Gpu	OpSys	Price	Memory(GB)	Ram(GB)	Weight(KG)	SSD	HDD
0	Apple	13.3	Intel Core i5 2.3GHz	Intel Iris Plus Graphics 640	macOS	71378	128	8	1.37	128	0
1	Apple	13.3	Intel Core i5 1.8GHz	Intel HD Graphics 6000	macOS	47895	128	8	1.34	128	0
2	HP	15.6	Intel Core i5 7200U 2.5GHz	Intel HD Graphics 620	No OS	30636	256	8	1.86	256	0
3	Apple	15.4	Intel Core i7 2.7GHz	AMD Radeon Pro 455	macOS	135195	512	16	1.83	512	0
4	Apple	13.3	Intel Core i5 3.1GHz	Intel Iris Plus Graphics 650	macOS	96095	256	8	1.37	256	0

In [ ]:

Out [ ]:

```
HDD
0    1273
Name: count, dtype: int64
```

In [122]:

```
data.head()
```

Out [122]:

	Company	Inches	Cpu	Gpu	OpSys	Price	Memory(GB)	Ram(GB)	Weight(KG)	SSD	HDD
0	Apple	13.3	Intel Core i5 2.3GHz	Intel Iris Plus Graphics 640	macOS	71378	128	8	1.37	128	0
1	Apple	13.3	Intel Core i5 1.8GHz	Intel HD Graphics 6000	macOS	47895	128	8	1.34	128	0
2	HP	15.6	Intel Core i5 7200U 2.5GHz	Intel HD Graphics 620	No OS	30636	256	8	1.86	256	0
3	Apple	15.4	Intel Core i7 2.7GHz	AMD Radeon Pro 455	macOS	135195	512	16	1.83	512	0
4	Apple	13.3	Intel Core i5 3.1GHz	Intel Iris Plus Graphics 650	macOS	96095	256	8	1.37	256	0

In [128]:

```
data.head()
```

Out [128]:

	Company	Inches	Cpu	Gpu	OpSys	Price	Memory(GB)	Ram(GB)	Weight(KG)	SSD
0	Apple	13.3	Intel Core i5 2.3GHz	Intel Iris Plus Graphics 640	macOS	71378	128	8	1.37	128
1	Apple	13.3	Intel Core i5 1.8GHz	Intel HD Graphics 6000	macOS	47895	128	8	1.34	128
2	HP	15.6	Intel Core i5 7200U 2.5GHz	Intel HD Graphics 620	No OS	30636	256	8	1.86	256
3	Apple	15.4	Intel Core i7 2.7GHz	AMD Radeon Pro 455	macOS	135195	512	16	1.83	512
4	Apple	13.3	Intel Core i5 3.1GHz	Intel Iris Plus Graphics 650	macOS	96095	256	8	1.37	256

In [135]:

```
data.drop(['Memory(GB)'], axis=1, inplace=True)
```

In [136]:

```
data.head(5)
```

Out [136]:

	Company	Inches	Cpu	Gpu	OpSys	Price	Ram(GB)	Weight(KG)	SSD
0	Apple	13.3	Intel Core i5 2.3GHz	Intel Iris Plus Graphics 640	macOS	71378	8	1.37	128
1	Apple	13.3	Intel Core i5 1.8GHz	Intel HD Graphics 6000	macOS	47895	8	1.34	128
2	HP	15.6	Intel Core i5 7200U 2.5GHz	Intel HD Graphics 620	No OS	30636	8	1.86	256
3	Apple	15.4	Intel Core i7 2.7GHz	AMD Radeon Pro 455	macOS	135195	16	1.83	512
4	Apple	13.3	Intel Core i5 3.1GHz	Intel Iris Plus Graphics 650	macOS	96095	8	1.37	256

In [138]:

```
data["HDD"] = data["Price"].apply(lambda x: 1024 if x > 30000 else (512 if 20000 < x <= 30000 else 0))
```

In [139]:

```
data.head()
```

Out [139]:

	Company	Inches	Cpu	Gpu	OpSys	Price	Ram(GB)	Weight(KG)	SSD	HDD
0	Apple	13.3	Intel Core i5 2.3GHz	Intel Iris Plus Graphics 640	macOS	71378	8	1.37	128	1024
1	Apple	13.3	Intel Core i5 1.8GHz	Intel HD Graphics 6000	macOS	47895	8	1.34	128	1024
2	HP	15.6	Intel Core i5 7200U 2.5GHz	Intel HD Graphics 620	No OS	30636	8	1.86	256	1024
3	Apple	15.4	Intel Core i7 2.7GHz	AMD Radeon Pro 455	macOS	135195	16	1.83	512	1024
4	Apple	13.3	Intel Core i5 3.1GHz	Intel Iris Plus Graphics 650	macOS	96095	8	1.37	256	1024

In [152]:

```
data['Company'].value_counts()
```

Out [152]:

```
Company
Lenovo       290
```

```
Dell         287
HP           266
Asus         156
Acer         103
MSI           53
Toshiba       47
Apple         21
Samsung        9
Razer          7
Mediacom       7
Microsoft      6
Xiaomi         4
Vero           4
Chuwi          3
Google         3
LG             3
Huawei         2
Fujitsu        2
Name: count, dtype: int64
```

In [155]:

```
data.head()
```

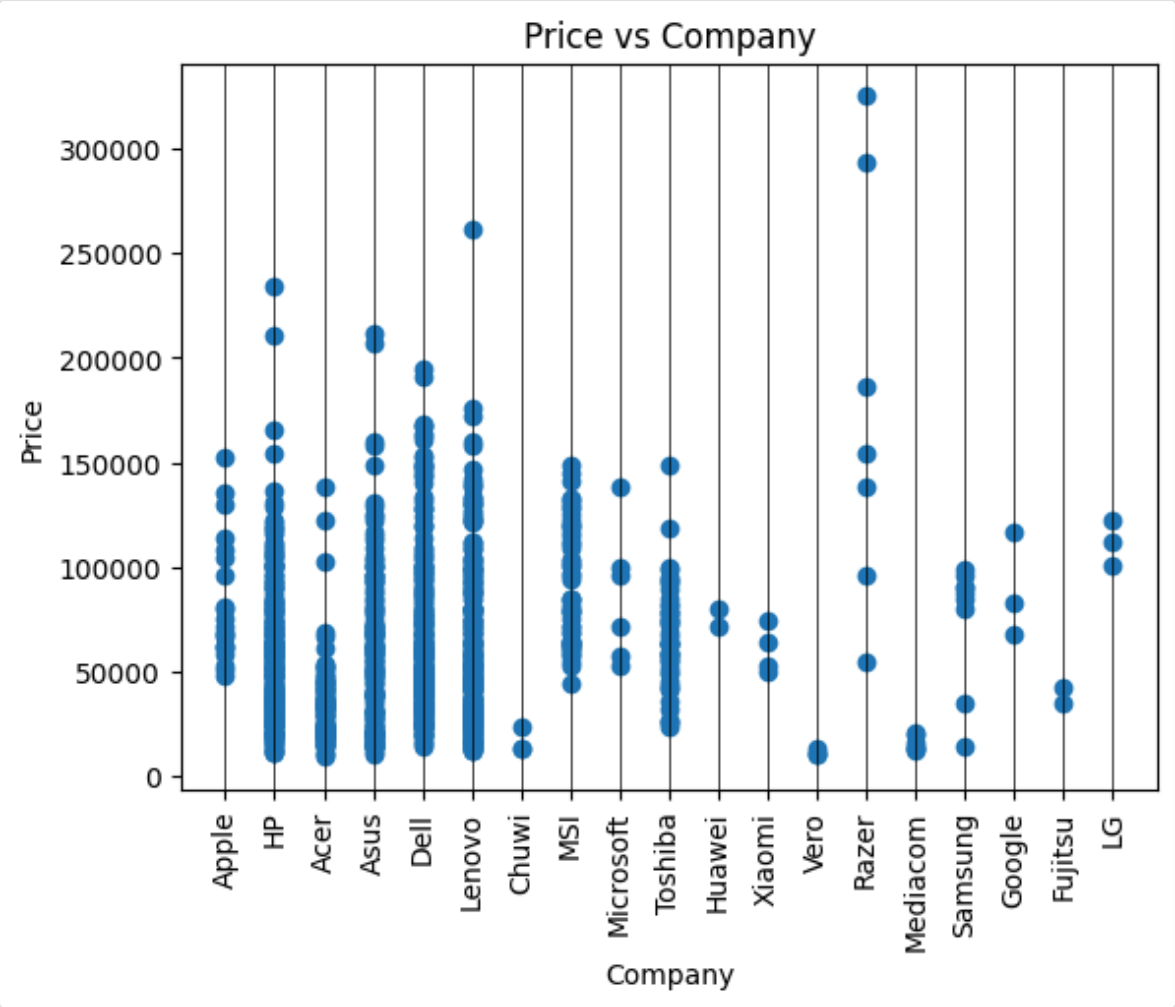
Out [155]:

	Company	Inches	Cpu	Gpu	OpSys	Price	Ram(GB)	Weight(KG)	SSD	HDD
0	Apple	13.3	Intel Core i5 2.3GHz	Intel Iris Plus Graphics 640	macOS	71378	8	1.37	128	1024
1	Apple	13.3	Intel Core i5 1.8GHz	Intel HD Graphics 6000	macOS	47895	8	1.34	128	1024
2	HP	15.6	Intel Core i5 7200U 2.5GHz	Intel HD Graphics 620	No OS	30636	8	1.86	256	1024
3	Apple	15.4	Intel Core i7 2.7GHz	AMD Radeon Pro 455	macOS	135195	16	1.83	512	1024
4	Apple	13.3	Intel Core i5 3.1GHz	Intel Iris Plus Graphics 650	macOS	96095	8	1.37	256	1024

In [174]:

```
x = data['Company']
y = data['Price']

plt.scatter(x, y)
plt.grid(axis='x',linewidth=0.5,color='Black')
plt.xticks(rotation=90)
plt.title('Price vs Company')
plt.xlabel('Company')
plt.ylabel('Price')
plt.show()
```

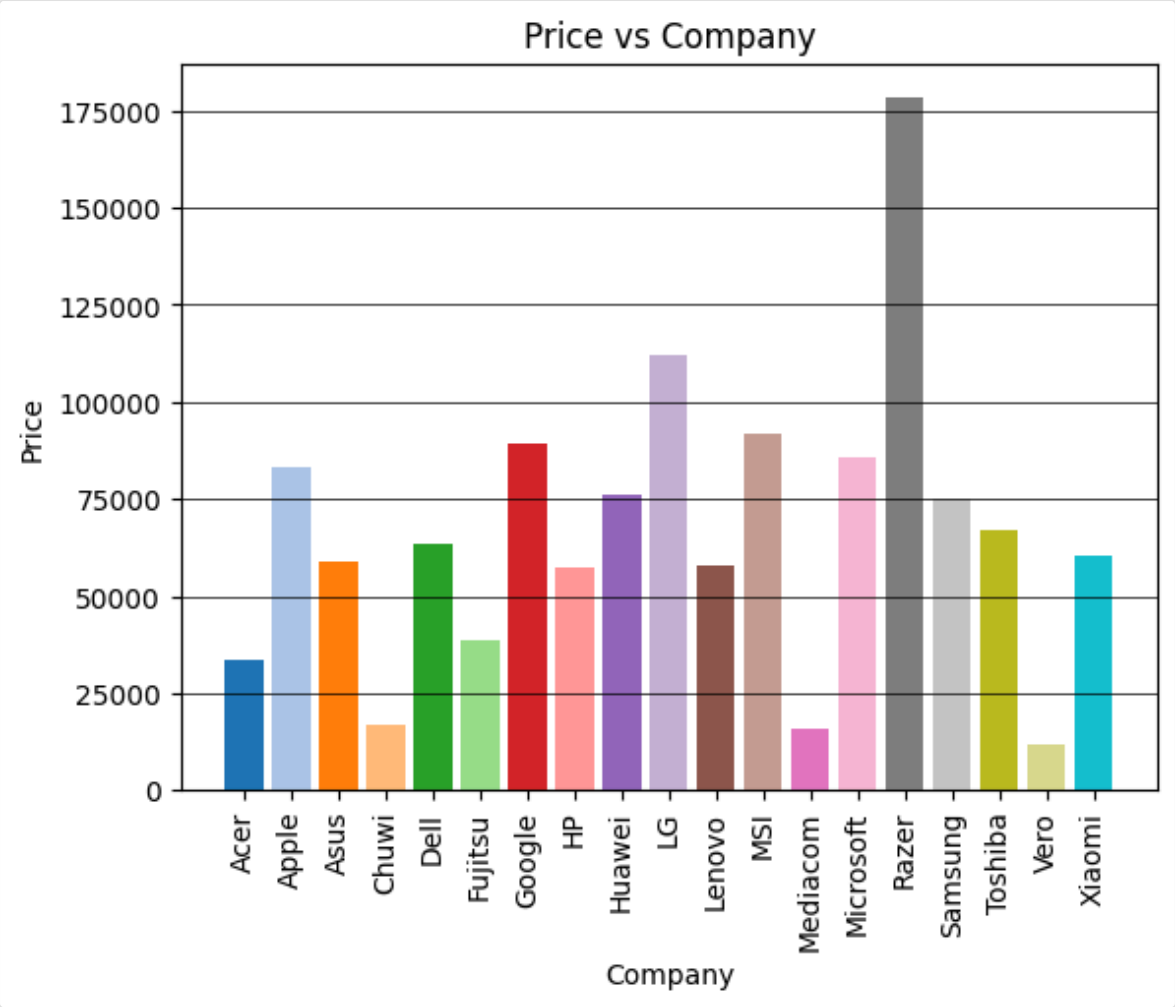
Out [174]:



In [194]:

```
counts = data.groupby('Company')['Price'].mean()

colors = plt.cm.tab20(range(len(counts)))

plt.bar(counts.index, counts.values, color=colors,)

plt.xlabel('Company')
plt.ylabel('Price')
plt.title('Price vs Company')
plt.xticks(rotation=90)

plt.grid(axis='y', linewidth=0.5, color='black')

plt.show()
```

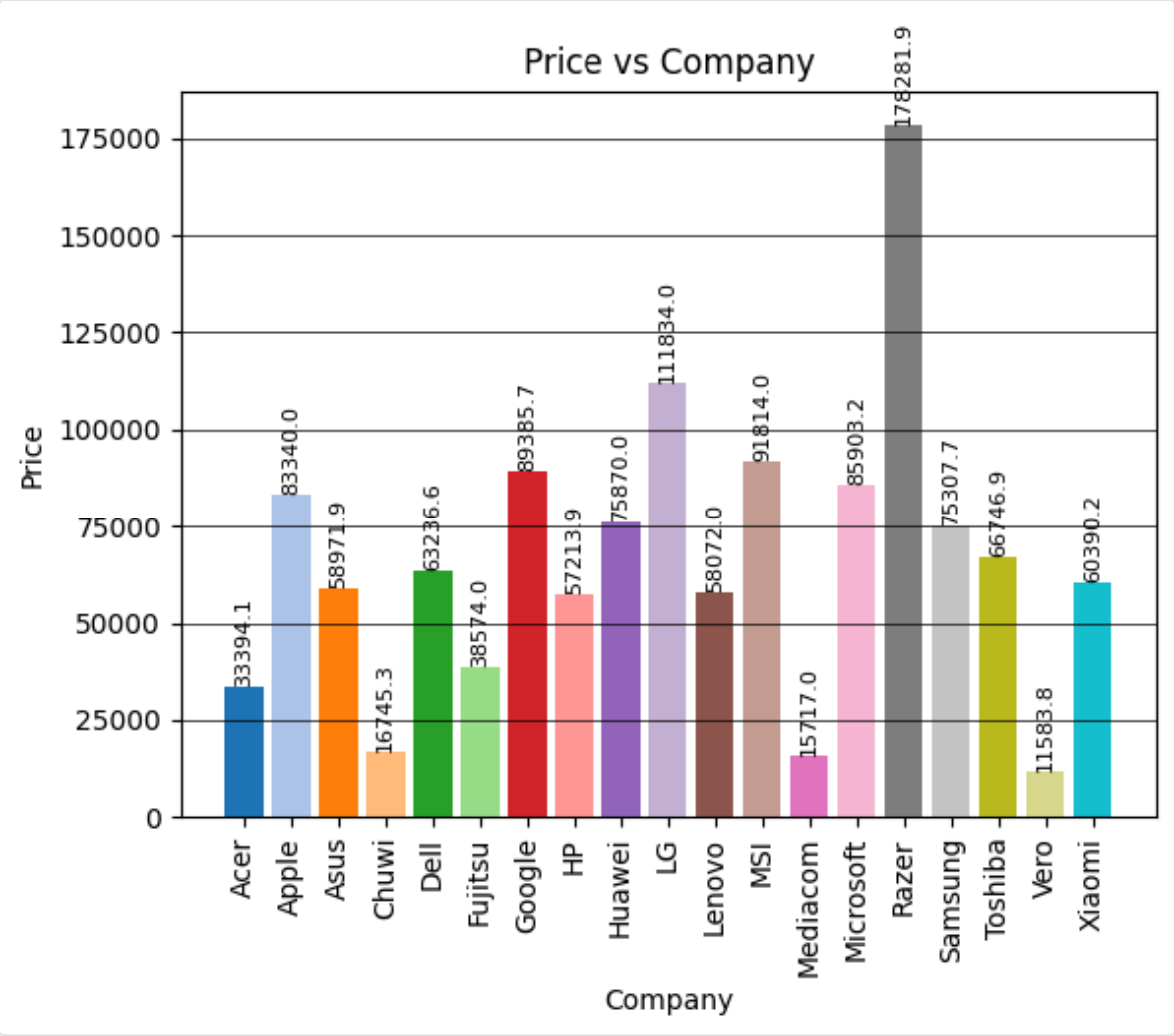
Out [194]:



In [196]:

```
counts = data.groupby('Company')['Price'].mean()

colors = plt.cm.tab20(range(len(counts)))

bars = plt.bar(counts.index, counts.values, color=colors)

plt.xlabel('Company')
plt.ylabel('Price')
plt.title('Price vs Company')
plt.xticks(rotation=90)

# 🎯 ADD VALUES ON TOP
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, yval, round(yval, 1),
             ha='center', va='bottom', fontsize=8, rotation=90)

plt.grid(axis='y', linewidth=0.5, color='black')

plt.show()
```

Out [196]:



In [197]:

```
data.head()
```

Out [197]:

	Company	Inches	Cpu	Gpu	OpSys	Price	Ram(GB)	Weight(KG)	SSD	HDD
0	Apple	13.3	Intel Core i5 2.3GHz	Intel Iris Plus Graphics 640	macOS	71378	8	1.37	128	1024
1	Apple	13.3	Intel Core i5 1.8GHz	Intel HD Graphics 6000	macOS	47895	8	1.34	128	1024
2	HP	15.6	Intel Core i5 7200U 2.5GHz	Intel HD Graphics 620	No OS	30636	8	1.86	256	1024
3	Apple	15.4	Intel Core i7 2.7GHz	AMD Radeon Pro 455	macOS	135195	16	1.83	512	1024
4	Apple	13.3	Intel Core i5 3.1GHz	Intel Iris Plus Graphics 650	macOS	96095	8	1.37	256	1024

In [201]:

```
x_train, x_test, y_train, y_test = train_test_split(data.drop('Price', axis=1), data['Price'], test_size=0.2, random_state=42)
```

In [202]:

```
x_train.head()
```

Out [202]:

	Company	Inches	Cpu	Gpu	OpSys	Ram(GB)	Weight(KG)	SSD	HDD
502	Lenovo	13.9	Intel Core i5 8250U 1.6GHz	Intel UHD Graphics 620	Windows 10	8	1.4	256	1024
416	Dell	13.3	Intel Core i3 6006U 2GHz	Intel HD Graphics 520	Windows 10	4	1.65	128	1024
162	Dell	15.6	Intel Core i5 8250U 1.6GHz	AMD Radeon 520	Windows 10	8	2.13	256	1024
667	HP	15.6	Intel Core i7 7500U 2.7GHz	AMD Radeon 530	Windows 10	8	2.1	1000	1024
792	Lenovo	13.3	Intel Core i7 7500U 2.7GHz	Intel HD Graphics 620	Windows 10	8	1.37	512	1024

In [203]:

```
y_train.head()
```

Out [203]:

```
502     85194
416     36709
162     38787
667     39693
792    104961
Name: Price, dtype: int64
```

In [205]:

```
y_test.shape
```

Out [205]:

```
(255,)
```